



MIMAN On-Board Computer

Data Sheet

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Document Information

Revision History

Version	Description	Editor	Date
R00	Initial release	Jinho Joung	03-Mar-2021
R01	FM Rev 1.0	Jinho Joung	01-Jun-2021
R02	FM Rev 1.0 (오타 및 내용 수정)	Jinho Joung	03-Jun-2021

1 Overview

NST OBC는 연세대학교에서 제작하는 큐브 위성으로 MIMAN에 장착될 온보드 컴퓨터이다. NST OBC는 ARM 기반의 400MHz 마이크로 프로세서를 탑재하였으며 사용자 요구에 맞춘 인터페이스 및 기능을 기반으로 MIMAN 임무를 성공적으로 수행할 수 있도록 제작되었다.

1.1 Features

1.1.1 Processor

Atmel ARM based Micro Processor Unit
Clock speed: 400 MHz

1.1.2 Memories & Storage

64-Kbyte ROM (internal)
32-Kbyte SRAM (internal)
256-Mbyte DDR2 RAM (external)
256-Mbyte NAND flash (external)
MicroSD slot (SDXC, Max 2TB)

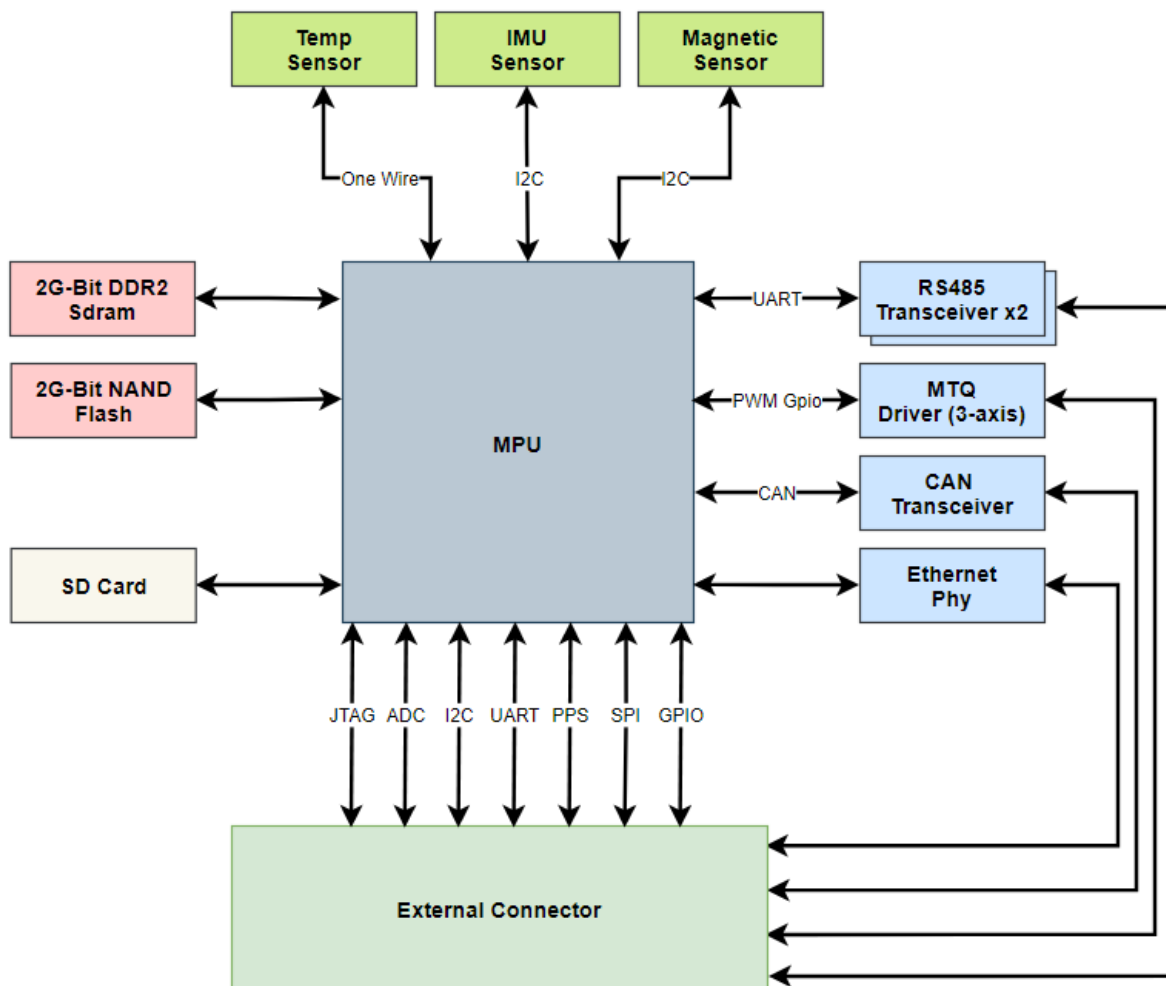
1.1.3 Peripherals

10/100 Mbps Ethernet MAC Controllers (EMAC) x 1
CAN Bus x 1, RS485 x 2, SPI x 1
Two-wire Interfaces (TWI) x 2, UART x 1, DUART x 1
ADC x 9, PPS x 1, GPIO x 5

1.1.4 Functions

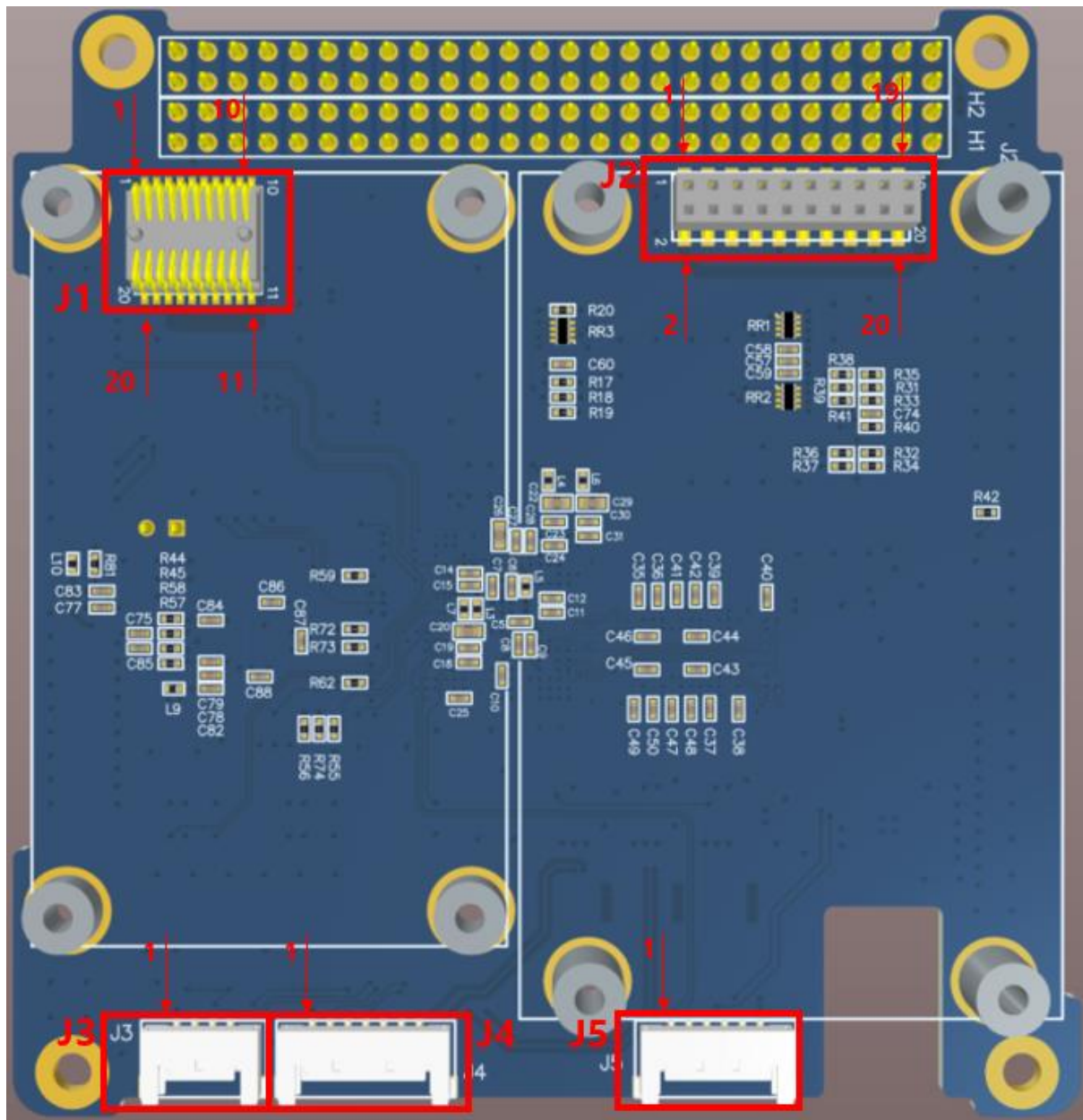
NovAtel OEM719 add-on slot x 1
Gomspace AX100 add-on slot x 1
MTQ controller x 3
On-board IMU Sensor (3-axis accelerometer ,3-axis gyroscope) x 1
On-board Magnetic Sensor (3-axis Magnetic) x 1
Temperature sensor x 1
External connector (User required)

2 Block Diagram

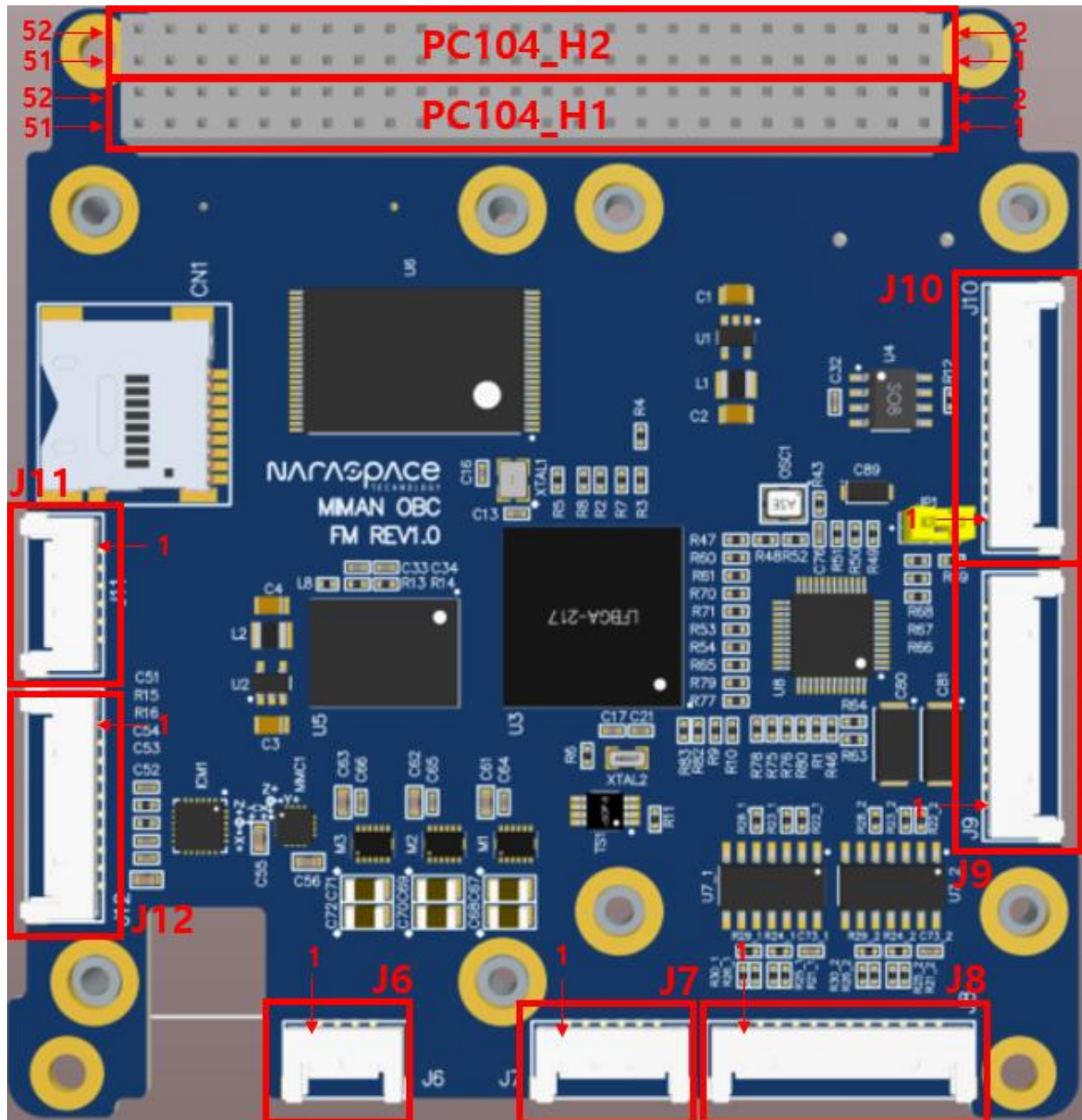


3 Hardware Layout

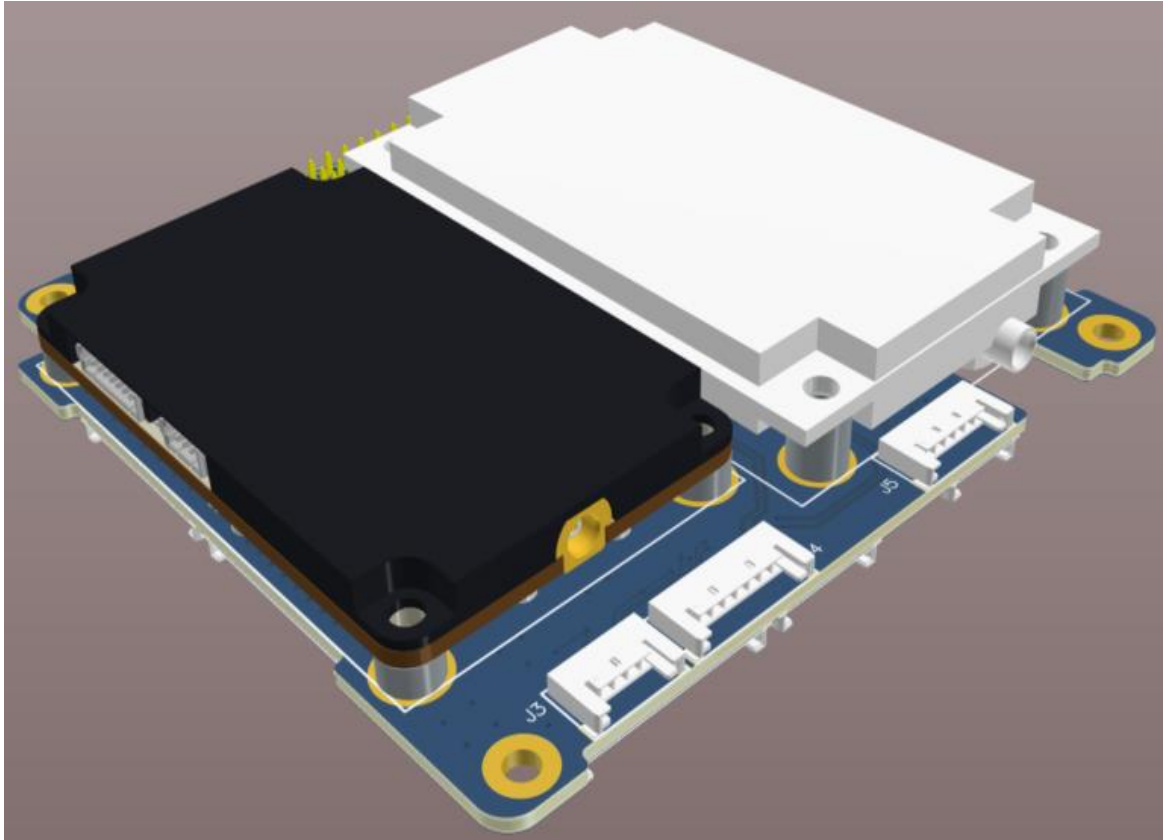
3.1 Top Layer



3.2 Bottom Layer



3.3 Application (with AX100 and OEM719)



4 Connector, Pinout Description

4.1 Top Connector

4.1.1 J1 (P/N : TLE-110-01-G-DV-A-TR, Samtec)

Description	J1		Description
GND AX100U	1	2	GND AX100U
3V3 SW2 AX100U	3	4	3V3 SW2 AX100U
I2C1 SCL	5	6	I2C1 SDA
CAN H	7	8	CAN L
NC	9	10	NC
NC	11	12	NC
NC	13	14	NC
NC	15	16	NC
3V3 SW2 AX100U	17	18	3V3 SW2 AX100U
GND AX100U	19	20	GND AX100U
GND: GND AX100U			
3V3: 3V3 SW2 AX100U			

4.1.2 J2 (P/N : TLE-110-01-G-DV-A-TR, Samtec)

Description	J2		Description
NC	1	2	3V3 SW3 OEM719
NC	3	4	NC
NC	5	6	NC
NC	7	8	NC
NC	9	10	GND OEM719
UART0 RXD	11	12	UART0 TXD
GND OEM719	13	14	NC
NC	15	16	GND OEM719
NC	17	18	GND OEM719
PPS OEM719	19	20	NC
GND: GND OEM719			
3.3V: 3V3 SW3 OEM719			

4.1.3 J3 (P/N : 504050-0491, Molex)

J3	Description
1	3.3V SW1 OBC
2	GND OBC
3	I2C2 SDA
4	I2C2 SCL
GND: GND OBC	
3.3V: 3.3V SW1 OBC	

4.1.4 J4 (P/N : 504050-0791, Molex)

J4	Description
1	SPI0 CS0
2	SPI0 SPCK
3	SPI0 MISO
4	SPI0 MOSI
5	CAN L
6	CAN H
7	GND OBC

4.1.5 J5 (P/N : 504050-0691, Molex)

J5	Description
1	GPIO2 (Release B)
2	GPIO1 (Release A)
3	GND UHF Ant.
4	I2C1 SDA
5	I2C1 SCL
6	5V SW2 UHF Ant.
GND: GND UHF Ant.	
5V: 5V SW2 UHF Ant.	

4.2 Bottom Connector

4.2.1 PC104_H1 (P/N : SSQ-126-01-G-D , Samtec)

Description	PC104_H1		Description
NC	1	2	NC
NC	3	4	GPIO4(STX_EN)
NC	5	6	NC
GPIO3(TR)	7	8	GPIO5(nFPGA_RESET)
NC	9	10	NC
SPI0 CS1	11	12	NC
NC	13	14	NC
NC	15	16	NC
NC	17	18	NC
NC	19	20	NC
SPI0 SPCK	21	22	GND STX
SPI0 MOSI	23	24	GND
NC	25	26	NC
GND	27	28	GND UHF Ant.
NC	29	30	NC
GND AX100U	31	32	GND OEM719
NC	33	34	NC
NC	35	36	NC
NC	37	38	NC
GND	39	40	GND OBC
I2C0 SDA	41	42	GND MTQ
I2C0 SCL	43	44	NC
NC	45	46	NC
5V SW1 MTQ	47	48	3V3 SW1 OBC
5V SW2 UHF Ant.	49	50	3V3 SW2 AX100U
5V SW3 STT	51	52	3V3 SW3 OEM719
GND: GND OBC, GND, GND MTQ, GND AX100U, GND STX, GND UHF Ant., GND OEM719 3.3V: 3V3 SW1 OBC, 3V3 SW2 AX100U, 3V3 SW3 OEM719 5V: 5V SW1 MTQ, 5V SW2 UHF Ant., 5V SW3 STT			

4.2.2 PC104_H2 (P/N : SSQ-126-01-G-D, Samtec)

Description	PC104_H2		Description
NC	1	2	NC
NC	3	4	NC
NC	5	6	NC
NC	7	8	NC
NC	9	10	NC
NC	11	12	NC
NC	13	14	NC
NC	15	16	NC
NC	17	18	NC
NC	19	20	NC
NC	21	22	NC
NC	23	24	NC
NC	25	26	NC
3V3 SW0 RAW Dig	27	28	NC
GND STX	29	30	GND
GND	31	32	GND
NC	33	34	NC
NC	35	36	NC
NC	37	38	NC
GND	39	40	GND
GND	41	42	GND
NC	43	44	GND RWA
NC	45	46	VBATT0 RWA
NC	47	48	NC
NC	49	50	NC
NC	51	52	NC
GND: GND RWA, GND STX 3.3V: 3V3 SW0 RAW Dig VBATT: VBATT0 RWA			

4.2.3 J6 (P/N : 504050-0491, Molex)

J6	Description
1	3.3V SW1 OBC
2	GND OBC
3	I2C2 SDA
4	I2C2 SCL
GND: GND OBC 3.3V: 3.3V SW1 OBC	

4.2.4 J7 (P/N : 504050-0691, Molex)

J7	Description
1	MTQ Z-
2	MTQ Z+
3	MTQ Y-
4	MTQ Y+
5	MTQ X-
6	MTQ X+

4.2.5 J8 (P/N : 504050-1291, Molex)

J8	Description
1	3V3 SW0 RWA Dig
2	GND RWA
3	5V SW3 STT
4	GND RWA
5	VBATT0 RWA
6	GND RWA
7	RS485 Y USART3
8	RS485 Z USART3
9	RS485 A USART0
10	RS485 B USART0
11	CAN H
12	CAN L
GND : GND RWA 3.3V: 3V3 SW0 RWA Dig 5V : 5V SW3 STT VBATT : VBATT0 RWA	

4.2.6 J9 (P/N : 504050-1291, Molex)

J9	Description
1	ETH1_TX_P
2	GND ETH1
3	ETH1_TX_N
4	GND ETH1
5	ETH1_RX_P
6	GND ETH1
7	ETH1_RX_N
8	GND ETH1
9	ETH1 LED1
10	EHT1 LED2
11	EARTH ETH1
12	ETH1 CT
GND: GND ETH1, EARTH ETH1	

4.2.7 J10 (P/N : 504050-1291, Molex)

J10	Description
1	JTAG TDI
2	JTAG TMS
3	JTAG TCK
4	JTAG RTCK
5	JTAG TDO
6	JTAG NTRST
7	CAN L
8	CAN H
9	DRXD
10	DRXD
11	3V3 SW1 OBC
12	GND OBC
GND: GND OBC	
3.3V: 3V3 SW1 OBC	

4.2.8 J11 (P/N : 504050-0691, Molex)

J11	Description
1	SPI0 SPCK
2	SPI0 MOSI
3	SPI0 MISO
4	SPI0 CS2
5	GND OBC
6	Analog PDD UHF Ant.

4.2.9 J12 (P/N : 504050-1091, Molex)

J12	Description
1	Analog Temp. Vout 4
2	Analog Temp. Vout 3
3	Analog Temp. Vout 2
4	Analog Temp. Vout 1
5	Analog Temp PDD 4
6	Analog Temp PDD 3
7	Analog Temp PDD 2
8	Analog Temp PDD 1
9	GND OBC
10	3V3 SW1 OBC
GND: GND OBC 3.3V: 3V3 SW1 OBC	

5 Recommended Electrical Characteristics

Parameter		Min	Typ	Max	Unit
Operating voltage		-	3.3	-	V
Current @ idle		-	130	-	mA
MTQ Controller	Power supply voltage(VM)	-	5	-	V
	Power supply voltage(VCC)	-	3.3	-	V
	Output current per channel	-1.5	-	1.5	A
	Control input voltage	-0.5	-	VCC+0.5	V
Gyroscope_[1]	Full-scale range	±250	-	±2,000	°/s
	ADC word length	-	16	-	bits
	Nonlinearity	-	±0.1	-	%
	Cross-axis sensitivity	-5	±2	+5	%
Accelerometer_[1]	Full scale range	±2	-	±16	g
	ADC word length	-	16	-	bits
	Nonlinearity	-	±0.5	-	%
	Cross-axis sensitivity	-5	±2	+5	%
Magnetometer_[1]	Field range(Each axis)	-	±8	-	G
	Output resolution	-	16	-	bits
	Total RMS noise	0.4	-	1.2	mG
CAN	Operating voltage	-	3.3	-	V
	Bit rate	-	250	1,000	kbps
RS485	Operating voltage	-	3.3	-	V
	Bit rate	-	115,200	-	Mbps
Ethernet	Operating voltage	-	3.3	-	V
	Bit rate	-	-	100	Mbps
I2C	Operating voltage	-	3.3	-	V
	Bit rate(Fast mode)	-	400	-	kbps
UART	Operating voltage	-	3.3	-	V
	Bit rate	-	38,400	-	bps
ADC	Operating voltage	0	-	3.3	V
	Resolution	-	10	-	bits

[1] 해당 기능의 상세 내용은 IC 데이터시트 참조 바람

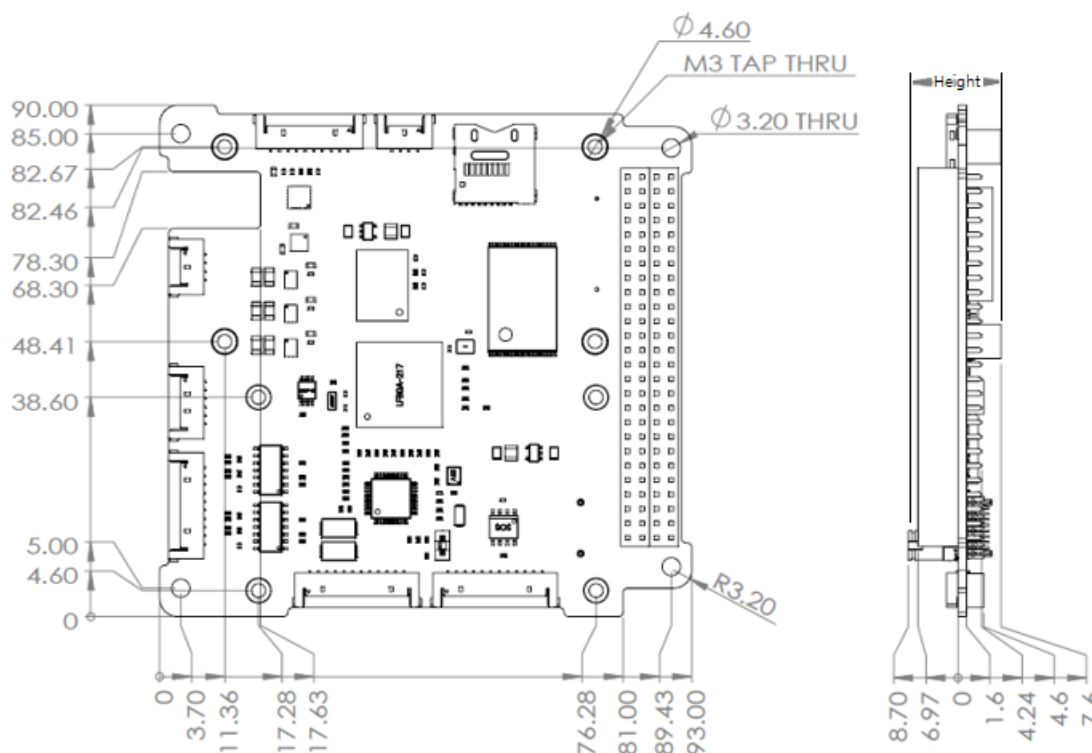
6 Absolute Electrical Characteristics

Parameter	Min	Max	Unit
Operating voltage (3.3V)	3.135	3.465	V
Operating voltage (5V)	TBD	6	V
Operating Temperature	-35	85	°C
Storage Temperature	-40	85	°C

7 Physical Characteristics

Parameter	Min	Unit
Mass	56	g
Dimensions	93 x 90 x 16.3	m ³

8 Mechanical Drawing



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9 Connector Information

	PCB Mount Connector		Wire Connector
Description	PCB Reference	Part Name	Part Name
12-Pin Connector	J8, J9, J10	504050-1291	504051-1201
10-Pin Connector	J12	504050-1091	504051-1001
7-Pin Connector	J4	504050-0791	504051-0701
6-Pin Connector	J5, J7, J11	504050-0691	504051-0601
4-Pin Connector	J3,J6	504050-0491	504051-0401
20-Pin Connector	J2	TLE-110-01-G-DV-A-TR	-
20-Pin Connector	J1	FSI-110-03-G-D-AD	-
PC104	H1, H2	SSQ-126-01-G-D	-